

Ethical AI Model Design Challenge

AI Application	AI System for Detecting Fake and Fraudulent Drug Prescriptions
Problem Statement	Prescription fraud is a growing problem that most people do not talk about. Doctors over-prescribe opioids and controlled substances under pressure or for profit, patients forge prescriptions to obtain drugs illegally, and pharmacies sometimes dispense without proper verification. The damage is real — addiction, black-market drug supply, and unnecessary deaths. This project proposes an AI system that analyses prescription data in real time to detect suspicious patterns — whether a single doctor is prescribing abnormally high doses, whether the same patient is collecting prescriptions from multiple clinics, or whether a document itself shows signs of being forged. It does not punish anyone automatically. It flags anomalies so that human investigators can follow up.
Data Required	• Prescription records — drug name, dosage, quantity, date, prescribing doctor ID, patient ID, pharmacy ID • Doctor profiles — specialisation, average prescription volume, historical prescribing behaviour • Patient history — frequency of prescription collection, number of clinics visited, prior drug-related flags • Pharmacy dispensing logs — cross-referenced with prescriptions to detect mismatches • Scanned prescription images — for forgery detection using document analysis (handwriting, stamp, format)
Expected Output	• Risk score per prescription — low / medium / high fraud likelihood • Flag type — e.g. "doctor over-prescribing," "duplicate prescription," "document forgery suspected," "pharmacy-prescription mismatch" • Visual anomaly map for scanned documents — highlighting inconsistencies in stamps, signatures, or formatting • Alert sent to a human review team — with a summary of why the prescription was flagged, not just that it was
Ethical Principles	1. Fairness — the system must not disproportionately flag prescriptions from lower-income patients or doctors working in under-resourced clinics, where prescription patterns may legitimately look unusual. 2. Privacy — prescription data is among the most sensitive personal information. Access must be strictly limited, all data anonymised where possible, and storage compliant with health data laws. 3. Transparency — every flag must come with a clear, human-readable reason. "The AI flagged it" is not acceptable. Investigators must understand exactly what triggered the alert.

	4. Accountability — no prescription should be blocked or a doctor suspended based on AI output alone. A qualified human must review every case before action is taken. 5. Safety & Reliability — a false positive that blocks a legitimate prescription for a cancer or chronic pain patient can cause serious harm. The system must be tested rigorously before deployment and monitored continuously after.
Ethical Risk	The most serious risk is wrongful flagging of legitimate prescriptions — what is called a false positive. Doctors in palliative care, oncology, or pain management routinely prescribe high doses of controlled substances because their patients genuinely need them. An AI trained mostly on general prescription data may not understand this context and flag these doctors unfairly. If the system is not carefully designed, it could end up discouraging doctors from prescribing what patients actually need — out of fear of being flagged. That would not be catching fraud. That would be creating a new kind of harm in the name of preventing one.
Mitigation Strategy	The model must be trained with medical specialty context built in — a palliative care doctor prescribing high-dose morphine should not be compared to a general practitioner doing the same thing. Specialty-specific baselines are essential. Every flag must go to a trained human investigator before any action is taken. The AI's role is to narrow down thousands of prescriptions to the ones worth a closer look — not to make decisions. A formal appeals process must exist for doctors and patients who believe they were wrongly flagged. And the system's false positive and false negative rates must be published and reviewed regularly by an independent body — not just by the organisation running it.